

Waiz Khan

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EDUCATION

Johns Hopkins University

Baltimore, MD

M.S.E. in Data Science

Aug. 2024 – Dec. 2026

• Coursework: Machine Translation, Artificial Intelligence, Deep Learning, Statistical Learning

GD Goenka University

Gurugram, India

B.Tech. in Computer Science & Engineering

Aug. 2019 – Nov. 2023

EXPERIENCE

Co-Founder & Founding Engineer

Apr. 2026 – Present

Arzaic LLC, Iris: an LLM agent platform for chronic care

Baltimore, MD

- Architected a staged LLM agent pipeline (gate → route → retrieve → generate → guard) with an embedding based semantic router, agentic retrieval tools over a multimodal patient record, and a safety judge that fails closed on every response
- Engineered the platform beneath it: a provider agnostic LLM client layer with streaming and cancellation, multimodal RAG that captions clinical artifacts before embedding them, and a versioned patient model that changes only through auditable patches
- Made evaluation a CI gate: a safety suite of 12 cases blocks every merge, and a nightly autonomous agent drafts care plan changes that a clinician reviews and approves

Research Assistant

Dec. 2025 – Present

Johns Hopkins University, Advisor: Prof. Philipp Koehn

Baltimore, MD

- Design and validate evaluation metrics for LLM representations: created an intrinsic multilinguality metric and tested it across 17 open models (Qwen3, Llama 3.1, OLMo-2, BLOOM) against 30+ preregistered predictions, confound controlled correlations, and held out model families
- Run the full experimental program on a Slurm GPU cluster, from representation extraction at scale to intervention studies spanning 6 training objectives and 3 seeds; authored the 35 page technical report, briefs, and slides behind an NSF proposal I am currently co-developing with Prof. Koehn
- Earlier phase: benchmarked 15+ decoder architectures and learning rules for hieroglyph to text translation on a 61k example corpus, including multimodal variants with ConvNeXt sign image encoders; best decoders reach BLEU-4 30.3, ten times the Transformer baseline

Sales Analyst Intern

Jan. 2023 – Apr. 2023

KS Aviation Pvt Ltd

Gurugram, India

- Automated month end revenue reporting across 6 regions with versioned queries and locked inputs, cutting closing cycle time 25%; demand forecasts on 50K+ rows and executive dashboards cut review time 20%

Research Assistant

Aug. 2022 – Nov. 2022

GD Goenka University, Advisor: Dr. Yogesh Kumar

Gurugram, India

- Studied chronic school absenteeism on the IHDS-II national survey of 40k+ students; logistic regression, KNN and random forest risk models reached 74% accuracy under fair validation
- Distilled 92 candidate factors down to the 27 strongest drivers so the findings stayed actionable for school administrators

Data Scientist Intern

May 2022 – Sep. 2022

EBMS Solutions Pvt Ltd

Chennai, India

- Engineered a churn prediction pipeline (XGBoost with time series cross validation) deployed to a production CRM with automated retraining; campaigns it informed contributed to a +15% retention lift at AUC 0.82
- Standardized a self serve analytics layer with curated cohort tables and data quality monitoring; reporting cycle time fell 30%

PROJECTS

AttendOpt | Python, CatBoost, DoWhy, PuLP

Jan. 2026

- Extended my 2022 research into an end to end pipeline on real IHDS-II survey data (47,027 students): an Optuna tuned gradient boosting ensemble reaches ROC-AUC 0.841 against a 0.699 logistic baseline, with Platt and isotonic calibration
- Validated the distance and income barrier causally with DoWhy (ATE 0.14 per SD of distance; placebo and random common cause refutations pass); an ILP allocator doubles expected impact against random allocation at a fixed budget under stated effect assumptions, stress tested by 10,000 Monte Carlo runs

ExoSAGE | Python, FastAPI, D3

Jul. 2026

- Point in time screening of NASA TOI transit candidates with false discovery rate control and walk forward backtesting, plus mixed integer telescope hour allocation under correlated risk; 11 test suites over checksummed data snapshots

Black-Scholes Pricer | Python, NumPy, SciPy, Streamlit

Jan. 2025

- Production grade European options pricer: Greeks, Newton-Raphson implied volatility solver with Brent fallback, and arbitrage diagnostics such as put call parity and butterfly spreads; 91+ tests in CI

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, C++, Swift, Bash

LLMs & Agents: agent pipelines and tool use, RAG and embedding retrieval, prompt engineering, agent evaluation harnesses and safety gating, streaming LLM APIs with provider agnostic clients, Hugging Face Transformers

ML: PyTorch, scikit-learn, XGBoost, CatBoost; transformers, CNNs; causal inference (DoWhy); experimental design, preregistered evaluation, time series cross validation

Systems & Data: Node.js/Express, Next.js, SQLite/PostgreSQL, pandas/NumPy/SciPy, Slurm HPC clusters, Git, Docker